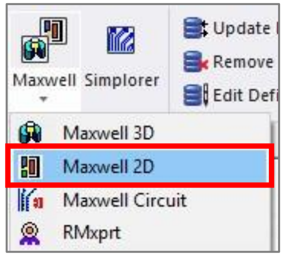
Workshop 06: Actuator Example

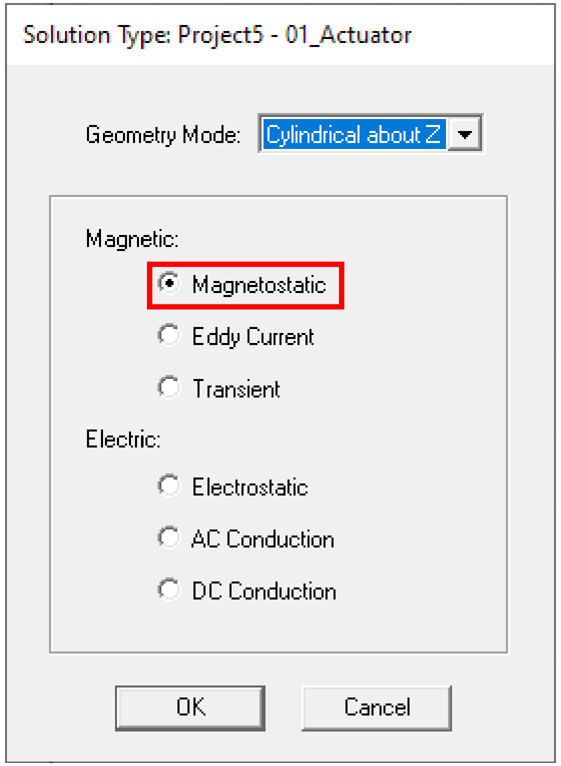
Overview

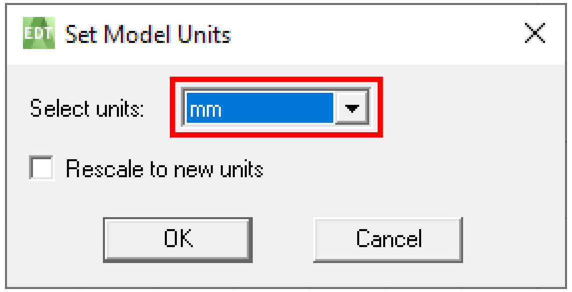
- **Actuator Example**
  - **This workshop will discuss how to use Maxwell in order to develop a complete analysis of a simple actuator in 2D Cylindrical symmetry**
  - **The following tasks will be performed:**
    - **Create the actuator model**
    - **Perform a Magnetostatic Analysis**
    - **Perform a Parametric Magnetostatic Analysis**
    - **Perform a Transient analysis**
    - **Analyze results**

Model Setup

- **Insert Design**
  - Select the menu item ***Project → Insert Maxwell 2D Design***, or click on the icon in drop down list Maxwell on panel Desktop
  - Name the Design “01_Actuator”
- **Set Solution Type**
  - Select the menu item ***Maxwell 2D → Solution Type***
  - Geometry Mode: ***Cylindrical about Z***
  - Choose ***Magnetic → Magnetostatic***
  - Click the OK button
- **Set Model Units**
  - Select the menu item ***Modeler → Units***
  - In Set Modeler Units window,
    - Select units: **mm** (millimeters)
    - Press the OK button

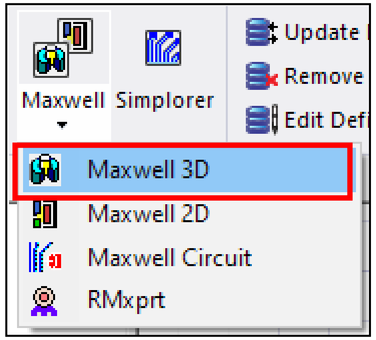


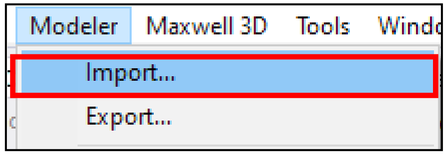


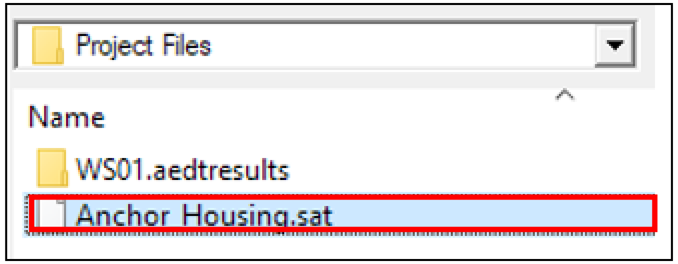


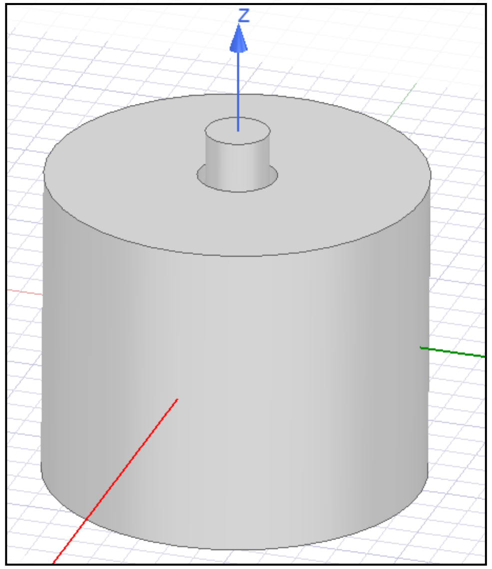
Create Model

- **Create the Housing and Anchor**
  - There are two ways of creating geometries in Maxwell
    - Importing .sat files or other CAD files to Maxwell
    - Drawing geometries from scratch
  - Creating Housing and Anchor by importing .sat file
    - Select the menu item *Project → Insert Maxwell 3D Design*, or click on the icon in drop down list Maxwell on panel Desktop
    - Select the menu item *Modeler → Import*
    - Choose *Anchor_Housing.sat → open*



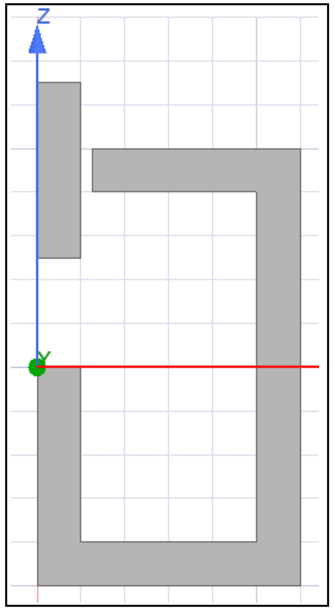


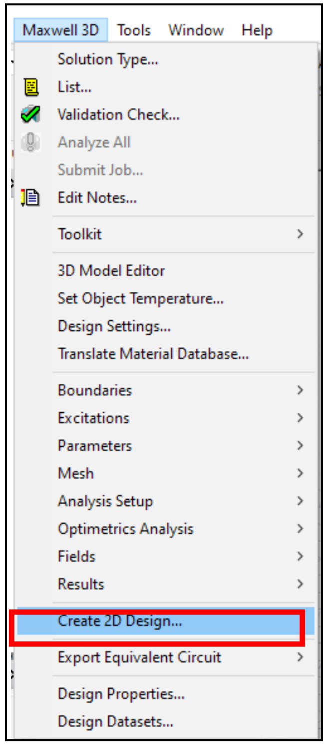


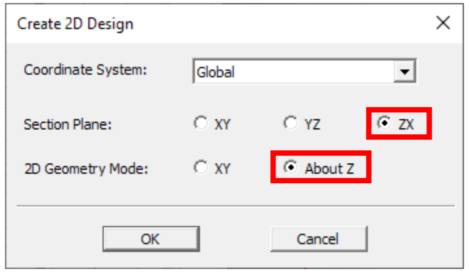


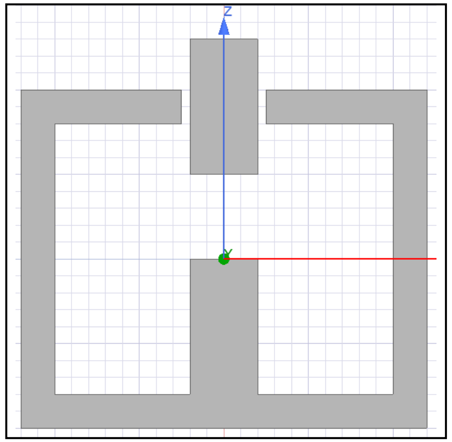
Create Model

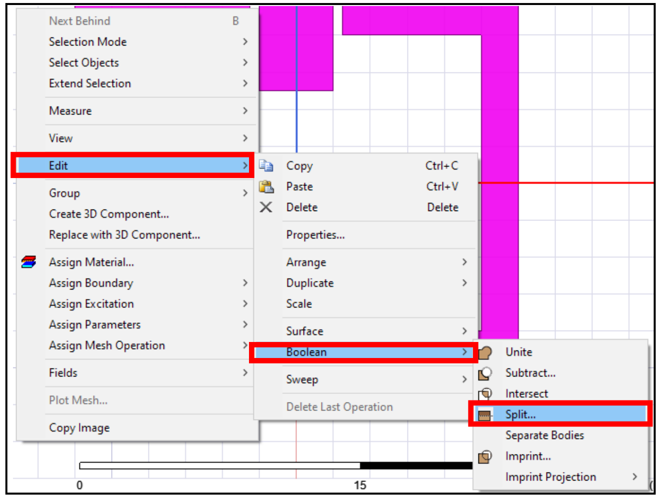
- **Create the Housing and Anchor**
  - Select the menu item *Maxwell → Create 2D Designs*
  - Select *ZX* for "Section Plane", and *About Z* for "2D Geometry Model"
  - Select both Housing and Anchor, *right click → Edit → Boolean → Split*
  - Select *YZ for* "Split using plane", keep result for *Positive side,* choose *Split all selected objects*, and check *Delete invalid objects created during operation*

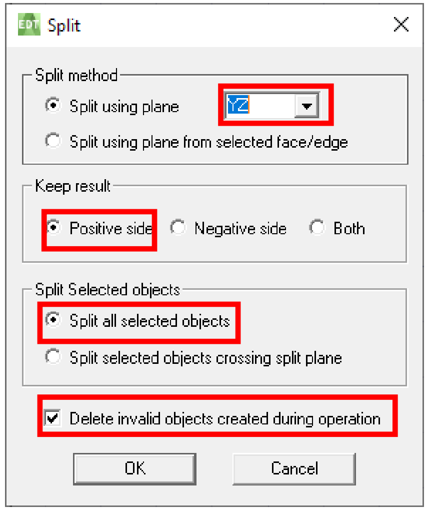










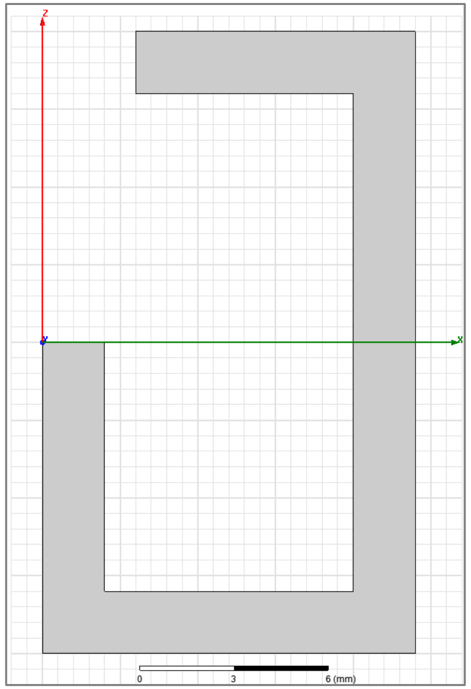


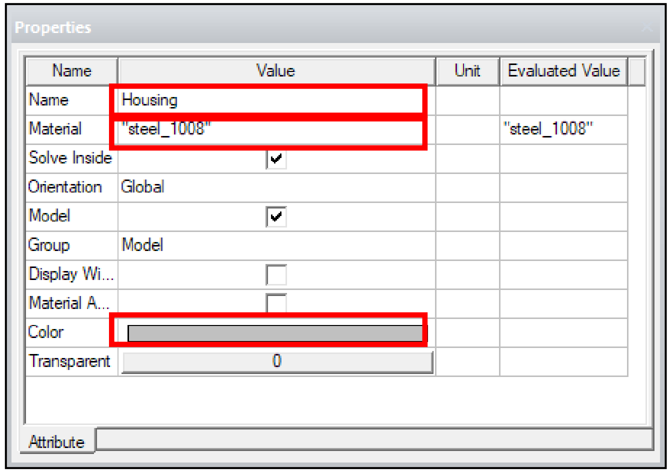
Create Model

- **Create the Housing**
  - Select the menu item ***Draw → Line***
    - Using the coordinate entry fields, enter the following points

X: 0, Y: 0, Z: 0,	*Press the Enter key*
X: 0, Y: 0, Z: -10,	*Press the Enter key*
X: 12, Y: 0, Z: -10,	*Press the Enter key*
X: 12, Y: 0, Z: 10,	*Press the Enter key*
X: 2.5, Y: 0, Z: 10,	*Press the Enter key*
X: 2.5, Y: 0, Z: 8,	*Press the Enter key*
X: 10, Y: 0, Z: 8,	*Press the Enter key*
X: 10, Y: 0, Z: -8,	*Press the Enter key*
X: 2, Y: 0, Z: -8,	*Press the Enter key*
X: 2, Y: 0, Z: 0,	*Press the Enter key*
X: 0, Y: 0, Z: 0,	*Press the Enter key twice*

  - Change the name of resulting sheet to Housing and color to Gray using the property window
  - Change the material of the sheet to Steel_1008

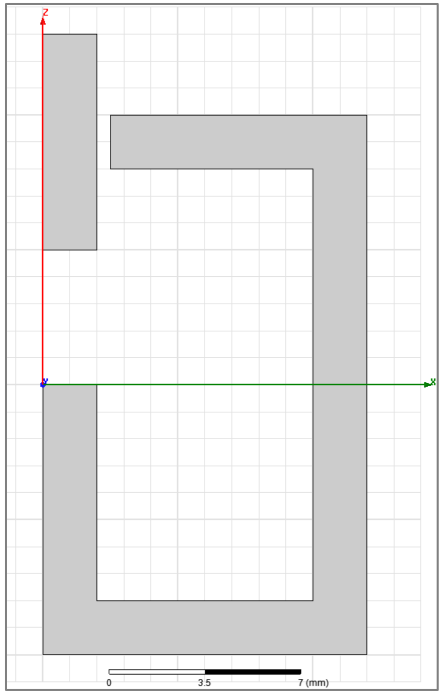


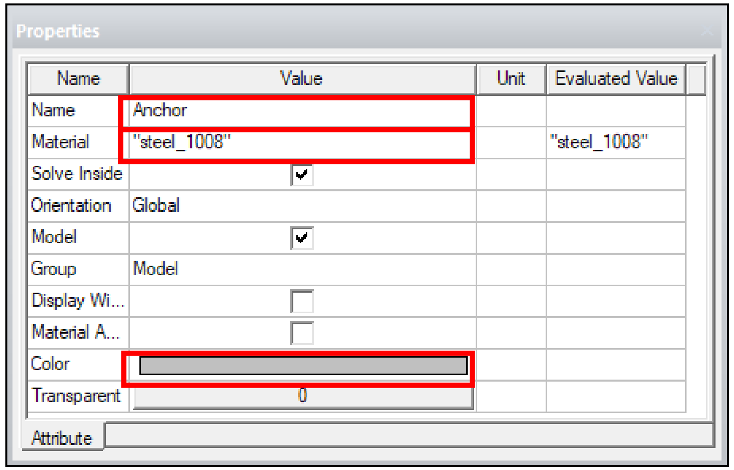


Create Model

- Create the Anchor
  – Select the menu item ***Draw → Rectangle***
    - Using the coordinate entry fields, enter the position of rectangle
      X: 0, Y: 0, Z: 13, *Press the Enter key*
    - Using the coordinate entry fields, enter the opposite corner
      dX: 2, dY: 0, dZ: -8, *Press the Enter key*

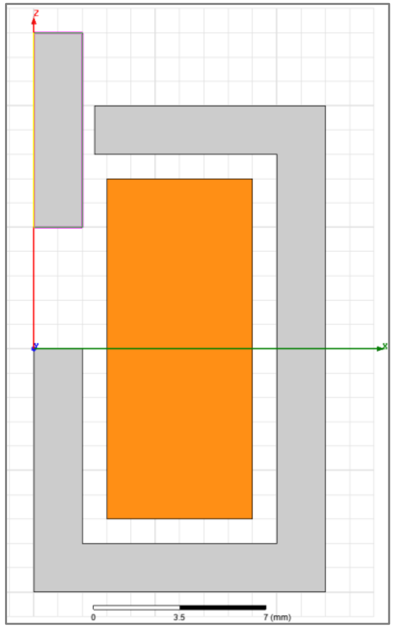
  – Change the name of resulting sheet to Anchor and color to Gray using the property window
  – Change the material of the sheet to Steel_1008

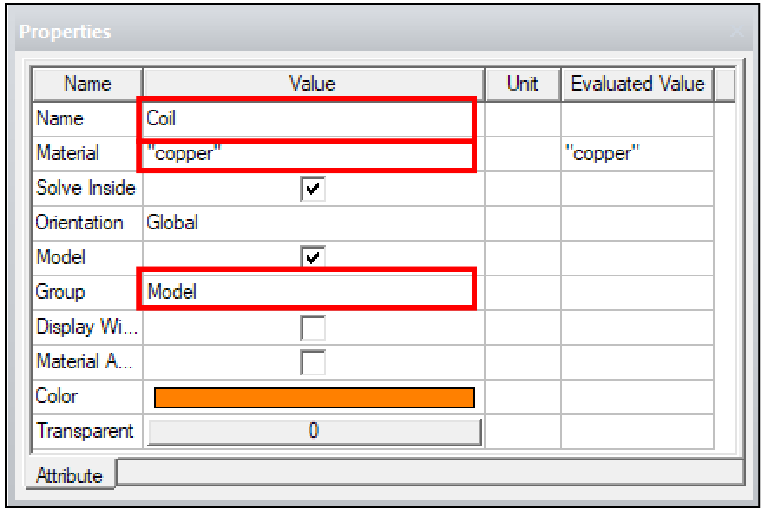




Create Model

- Create the Coil
  - Select the menu item *Draw → Rectangle*
    - Using the coordinate entry fields, enter the position of rectangle
      X: 3, Y: 0, Z: 7, *Press the Enter key*
    - Using the coordinate entry fields, enter the opposite corner
      dX: 6, dY: 0, dZ: -14, *Press the Enter key*
  - Change the name of resulting sheet to Coil and color to Orange using the property window
  - Change the material of the sheet to Copper



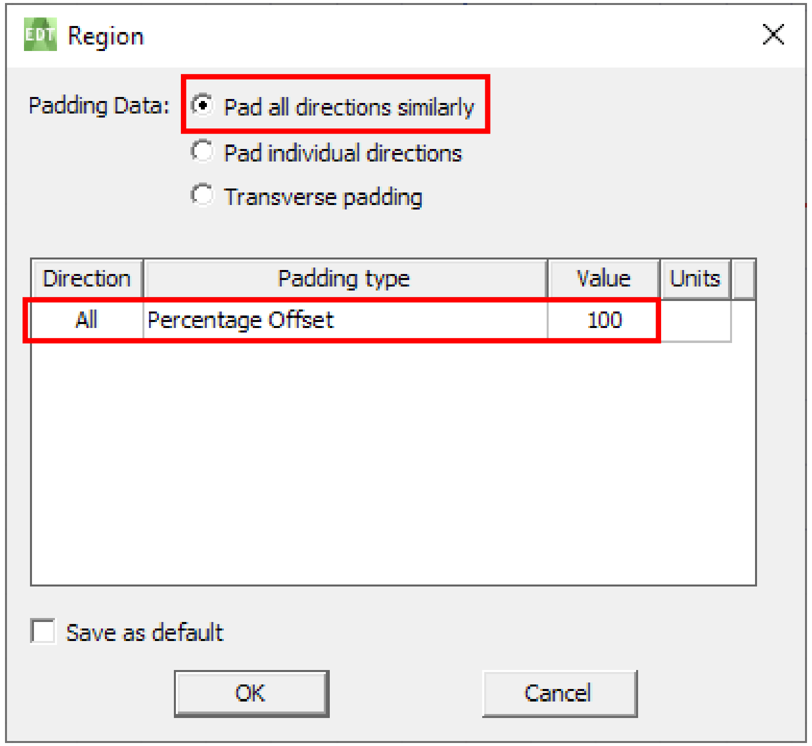


Define Region

- Create Simulation Region
  - Select the menu item *Draw → Region* or click on the icon 
  - In Region window
    - Pad all directions similarly: ☑ Checked
    - Padding Type: Percentage Offset
    - Value: 100
    - Press OK

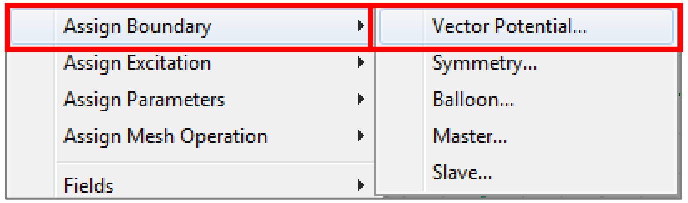
Note: the region extension in negative X direction is neglected due to RZ-symmetry about the Z-axis.

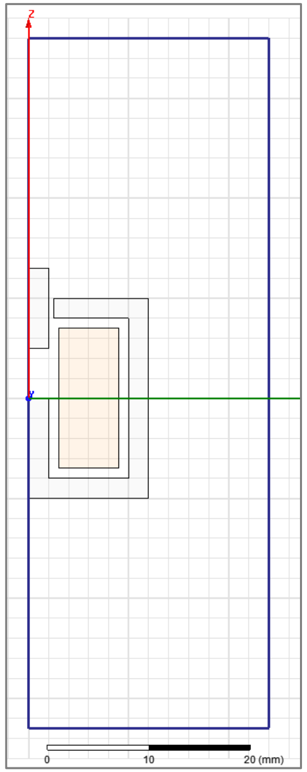
100 padding percentage is chosen to define the region size which should sufficiently capture the fringing flux, in certain cases, a larger sized region may be required.



Assign Boundary Conditions and Force Parameter

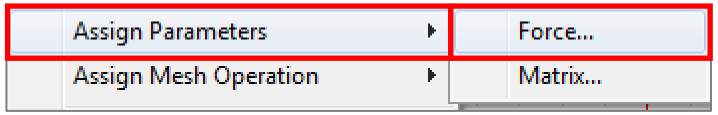
- **Assign Boundary Conditions to Region Edges**
  - Select the menu item *Edit → Select Mode → Edges*
  - Select the three edges of the Region
  - Select the menu item *Maxwell 2D → Boundaries → Assign → Vector Potential*
  - In Vector Potential Boundary window, Press OK





Note: the choice of vector potential boundary condition confines the magnetic flux to the size of the region

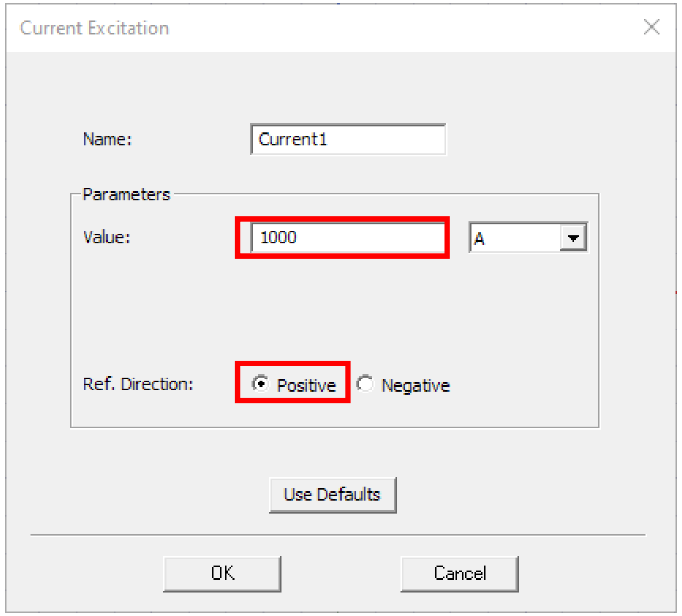
- **Assign Force Calculation**
  - Select the object Anchor from history tree
  - Select the menu item *Maxwell 2D → Parameters → Assign → Force*
  - In Force Setup window, Press OK

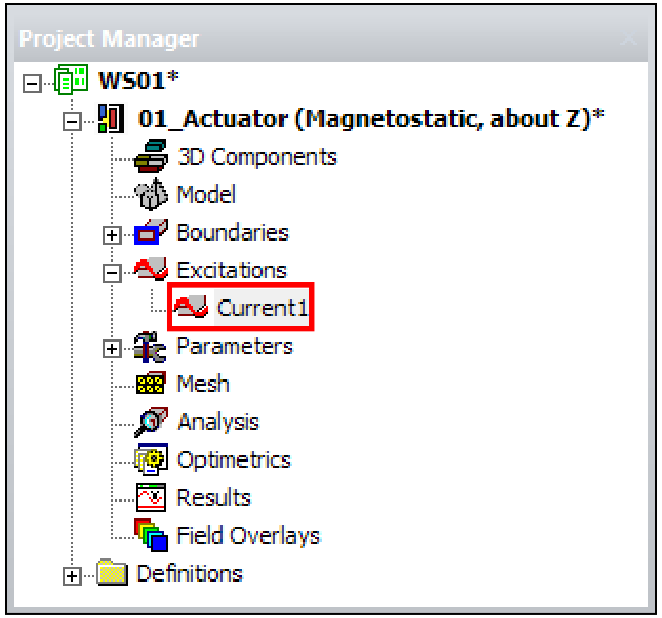


Assign Excitations

- **Assign Excitations**
  - Select the sheet Coil from history tree
  - Select the menu item *Maxwell 2D → Excitations → Assign → Current*
    - Name: *Current1*
    - Value: *1000*
    - Ref. Direction: *Positive*
    - In the Project Manager window under Excitations a new item called “Current1” is present

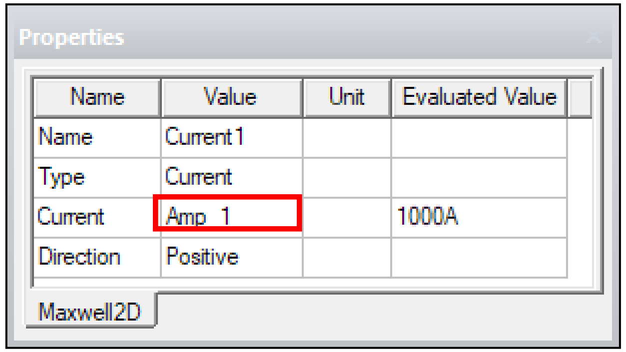
Note: A value of 1000 represents 1000 A-turns

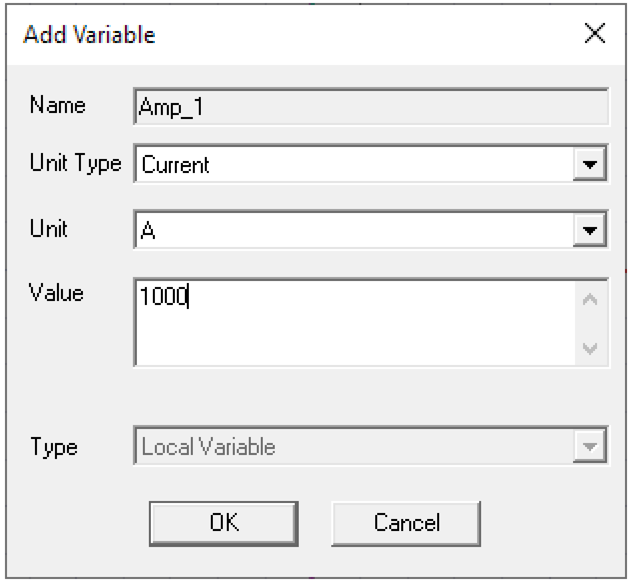




Prepare the Parametric analysis

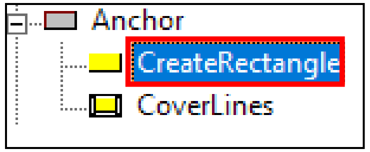
- **Modify Excitations**
  - Select the just created Excitation Current1
  - In the property window, change the current value from 1000 to Amp_1
  - The Add Variable window pops-up with all the fields already fulfilled. Press Ok
  - The new local variable Amp_1 has been added to 01_Actuator Design

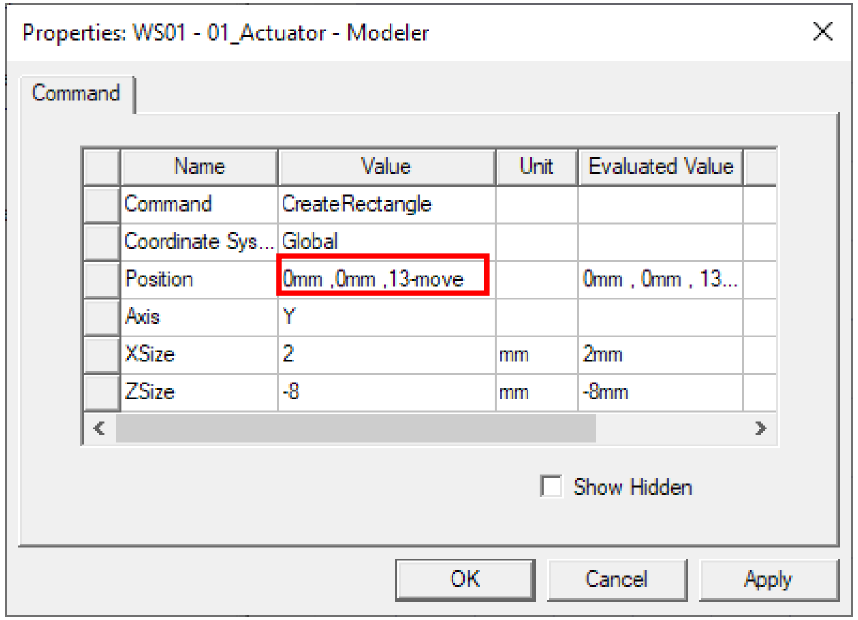


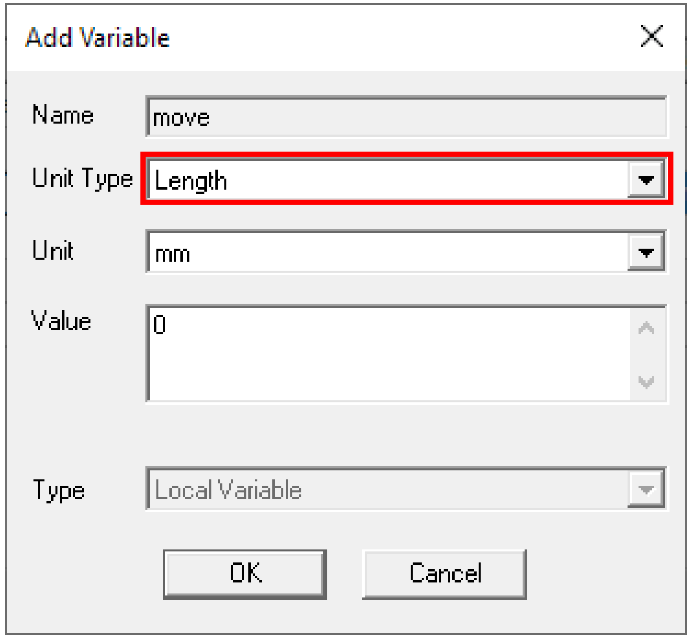


Prepare the Parametric analysis

- **Modify Anchor Position**
  - Select **CreateRectangle** from **Anchor** from the history tree
  - In the property window, change the Z-Position from 13 mm to **13mm - move**
  - The **Add Variable window** pops-up.
    - Unit Type: Length
    - Press OK
  - The new local variable **move** has been added to 01_Actuator Design

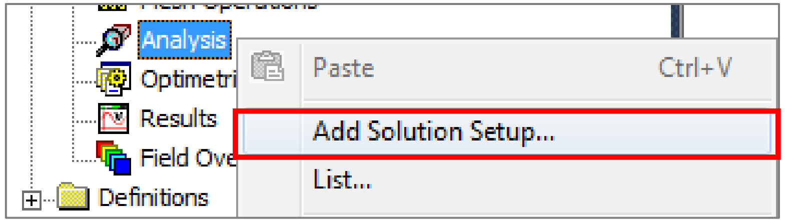


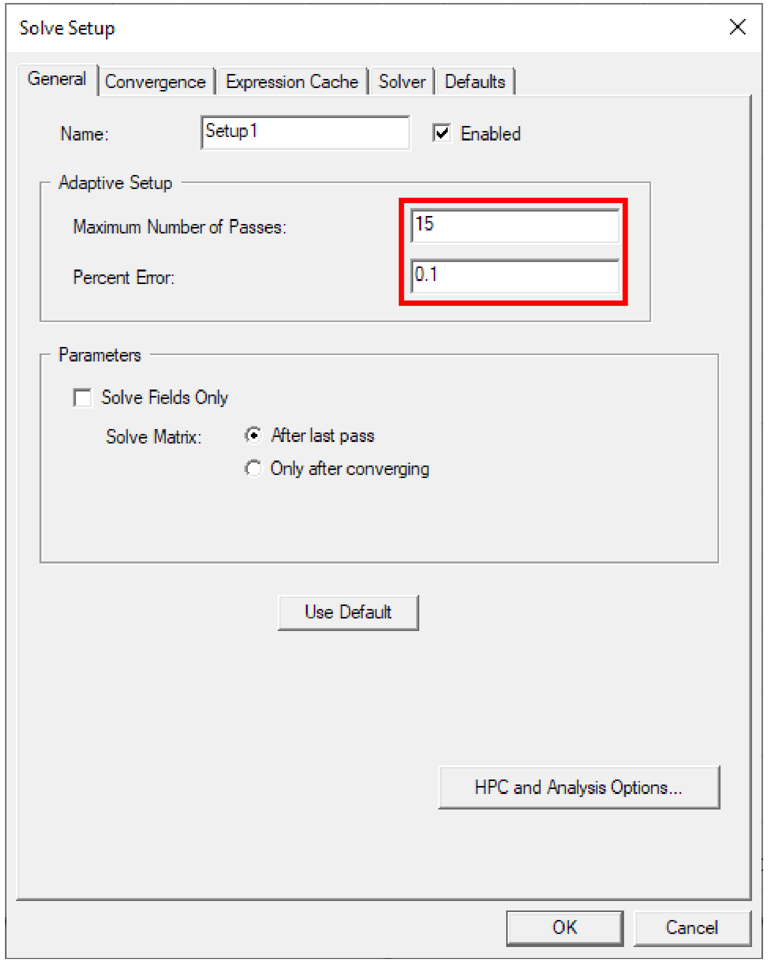


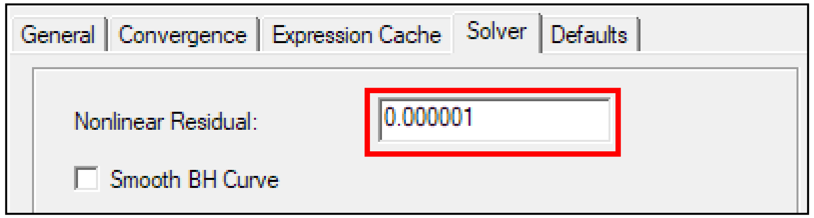


Analyze

- **Create an analysis setup:**
  - In Project Manager window *RMB on Analysis → Add Solution Setup*
  - Solution Setup Window:
    - General Tab
    - Maximum Number of Passes: 15
    - Percent error: 0.1%
    - Solver Tab
    - NonLinear Residual: 1e-6
    - Press OK
- **Start the solution process:**
  - In Project Manager window *RMB on Setup1 → Analyze*



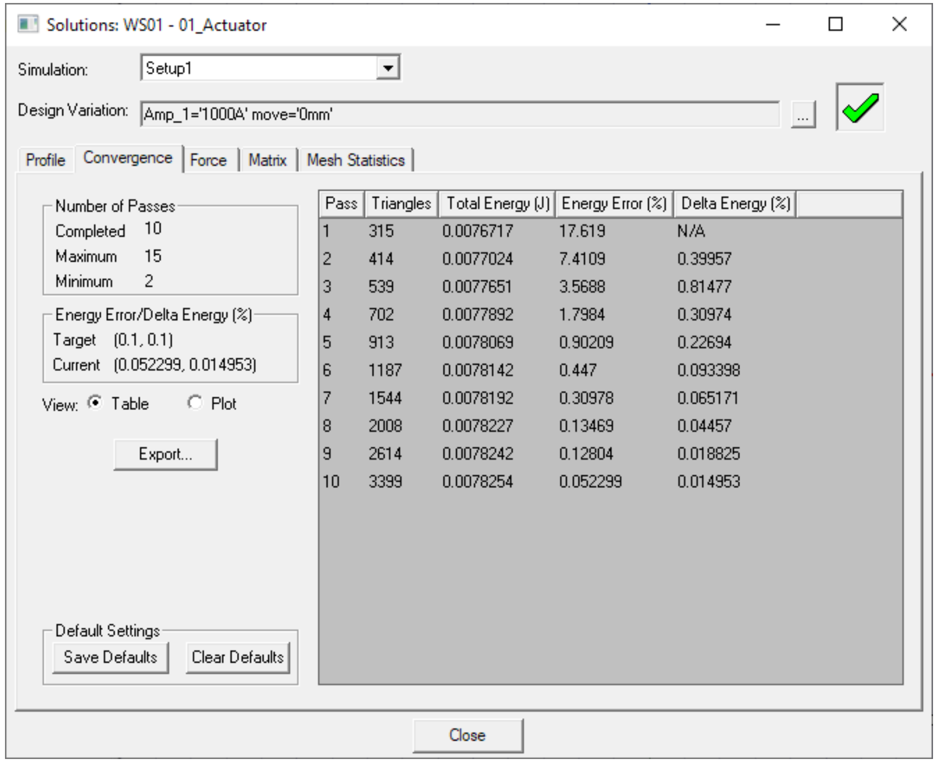




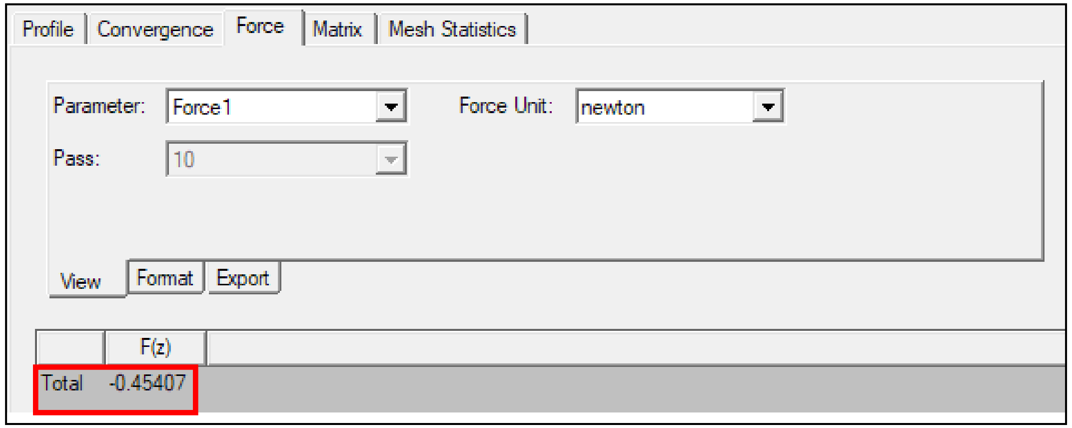
Ansys

Solution Data

- **View Solution Information**
  - In Project Manager window ***RMB on Results → Solution Data***
  - To View Convergence
    - Select the Convergence tab

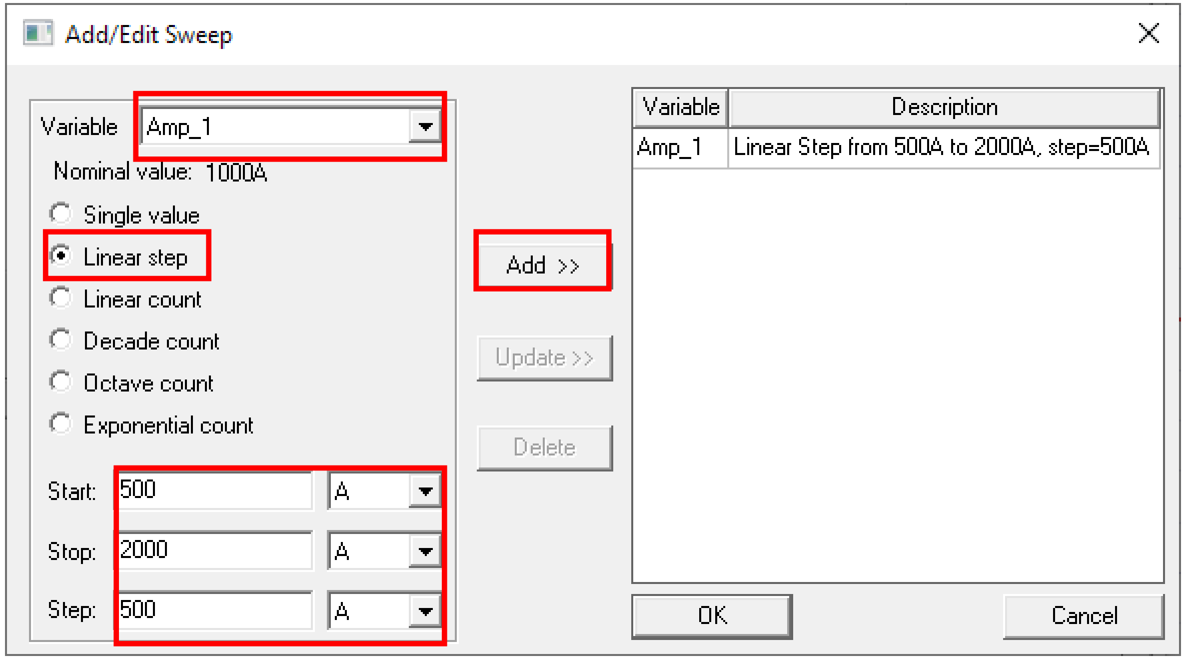


  - To View Calculated Force Value
    - Select the Force tab



Set the Parametric Analysis

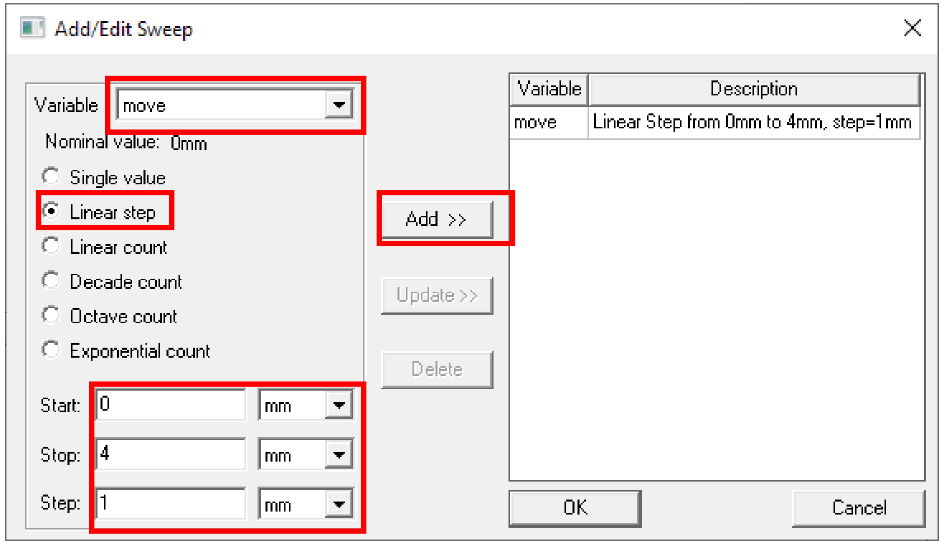
- **Create a Parametric analysis setup:**
  - **In Project Manager window** ***RMB on Optimetrics → Add → Parametric***
  - **Setup Sweep Analysis Window:**
    - **Press Add**
  - **Add/Edit Sweep Window:**
    - **Select Variable Amp_1**
    - **Linear Step: ☑ Checked**
    - **Start: 500 A**
    - **Stop: 2000 A**
    - **Step: 500 A**
    - **Press Add >>**
    - **Press OK**

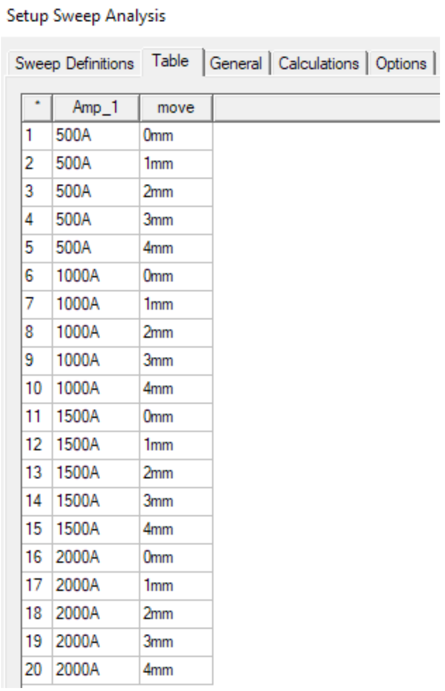


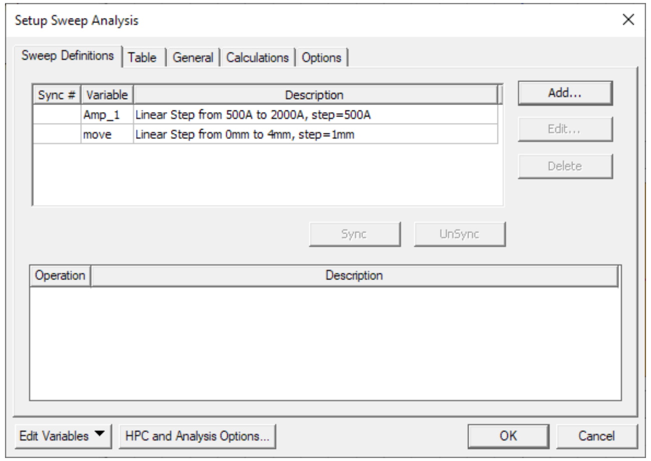
Ansys

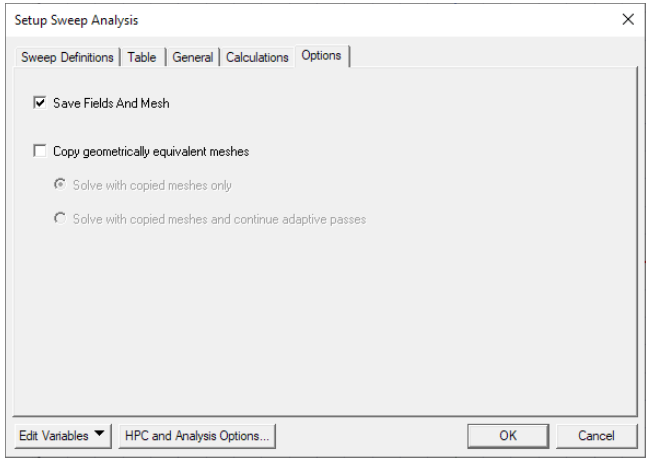
Set the Parametric Analysis

- Setup Sweep Analysis Window:
  - Press Add
- Add/Edit Sweep Window:
  - Select Variable move
  - Linear Step: ☑ Checked
  - Start: 0 mm
  - Stop: 4 mm
  - Step: 1 mm
  - Press Add >>
  - Press OK
- Setup Sweep Analysis Window:
- Options Tab
  - Save Fields And Mesh: ☑ Checked
  - Press OK
  - Press OK



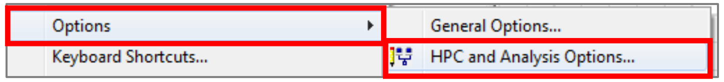


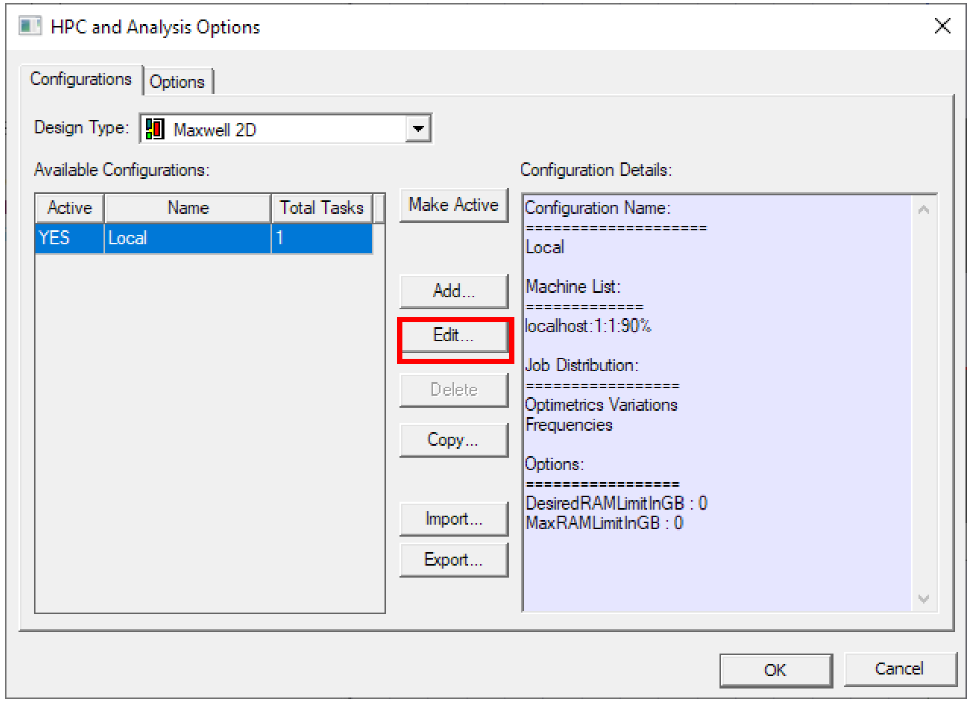


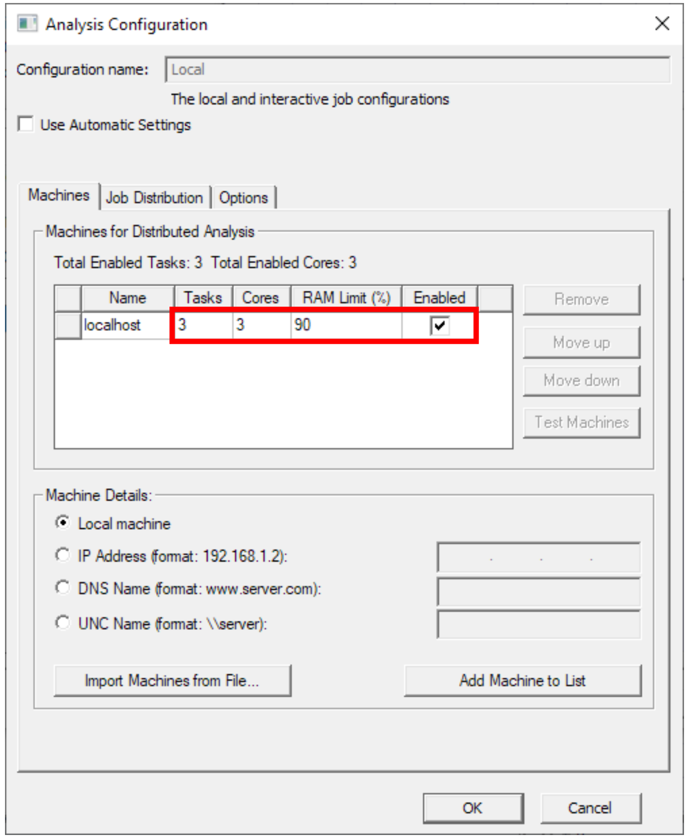


Settings DSO Analysis

- Activate DSO (Distributed Solver Option)
  - DSO is an additional option enabling the distribution of different sweeps through the processor's cores, speeding up Parametric analysis
  - Select menu *Tools→ Options→ HPC and Analysis Options*
  - HPC and Analysis Options Window:
    - Press Edit
    - Tasks: 3
    - Cores: 3
    - Press OK
    - Press OK



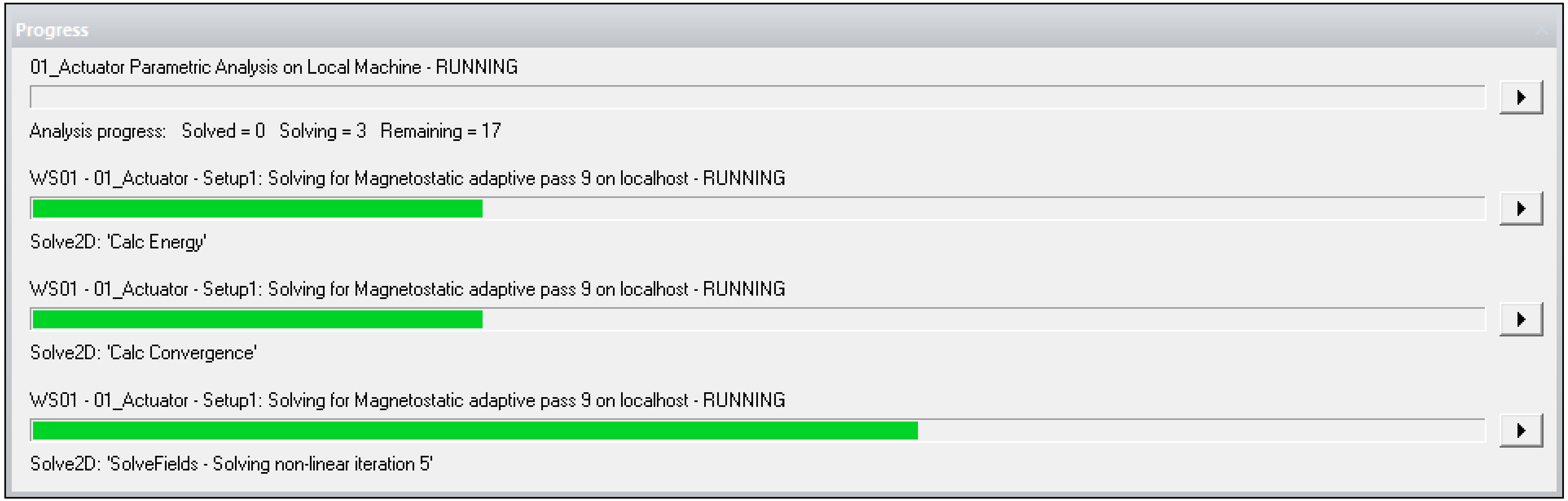




Ansys

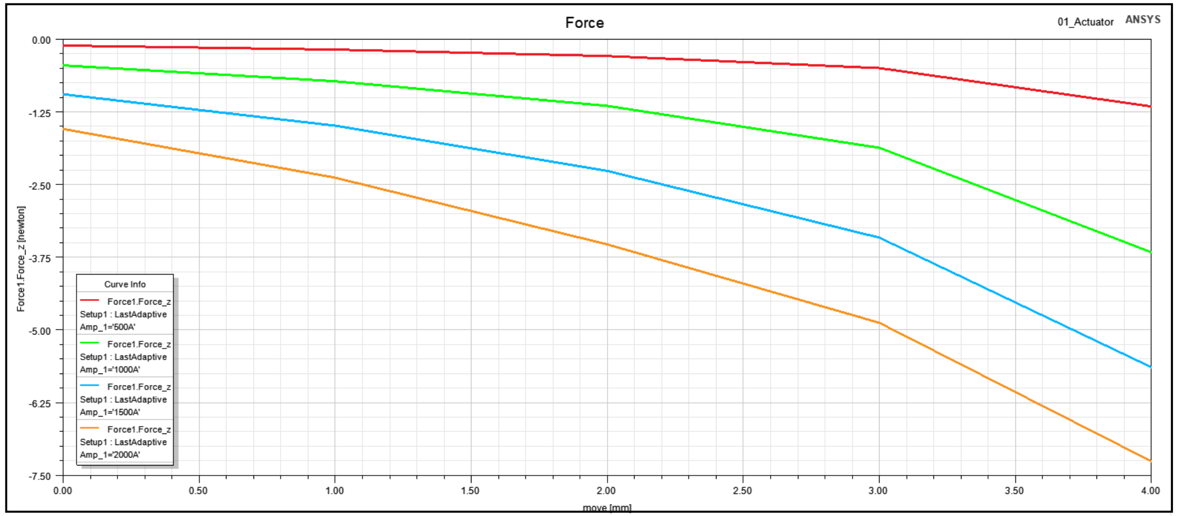
Analyze the Parametric Sweep

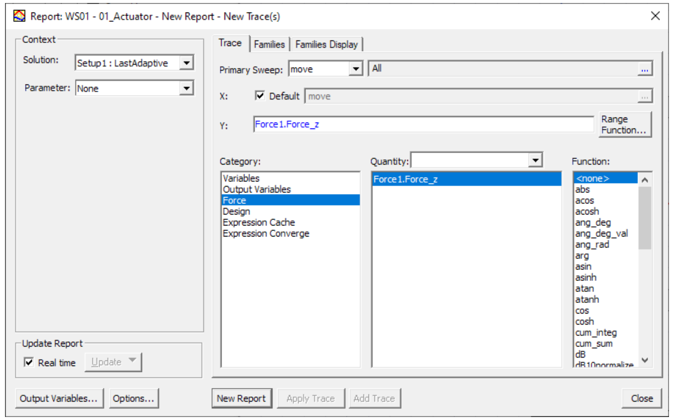
- **Launch the Parametric Analysis**
  - *RMB on ParametricSetup1 → Analyze*
  - In the Progress Window it is possible to see that 3 Sweeps at the same time are solved

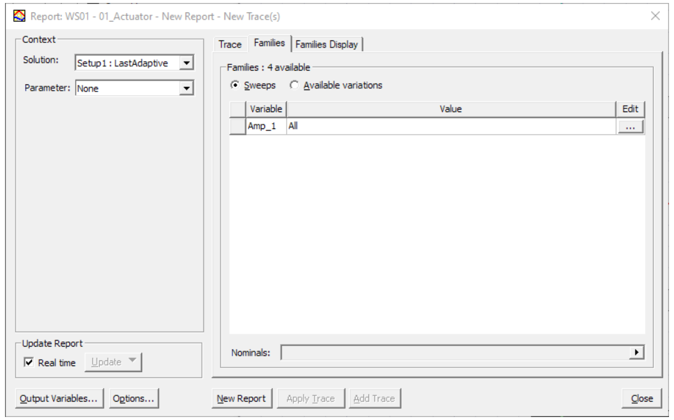


Create XY Plot

- **Plot the values of Force depending on Amp_1 and move Variables**
  - Select the menu item *Maxwell 2D → Results → Create Magnetostatic Report → Rectangular Plot*
  - In Report window (Trace Tab)
    - Primary sweep:
    - X : move
    - Category: Force
    - Quantity: Force1.Force_z
    - Families Tab
    - Amp_1: All
    - Select New Report



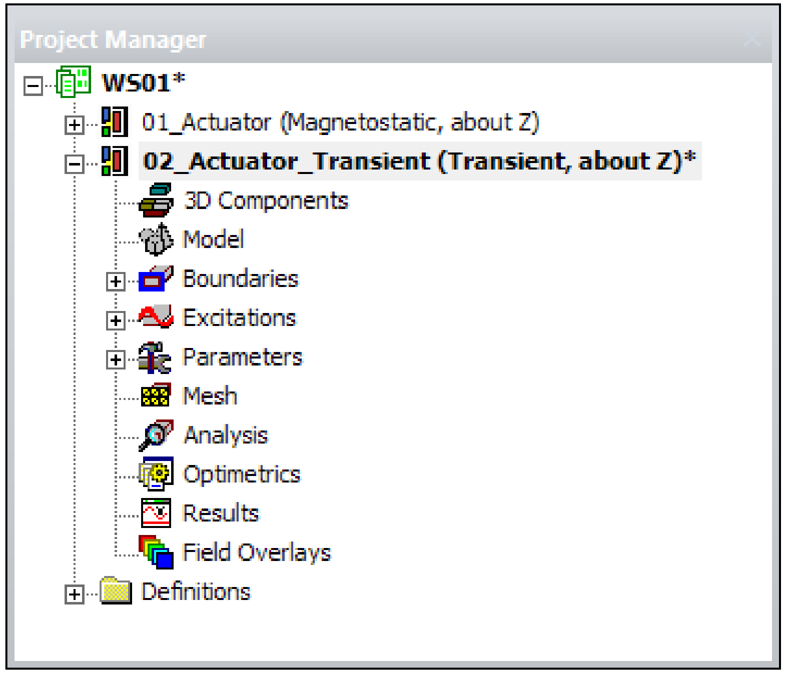




Transient Model

Create Transient Design

- **Copy the 01_Actuator design**
  - Select the Maxwell Design "01_Actuator"
  - Ctrl + C
  - Select the Maxwell Project
  - Ctrl + V
  - Click on the new Design name
  - Name the new Design "02_Actuator_Transient"
- **Change the Solution Type**
  - Select the menu item *Maxwell 2D → Solution Type*
  - Choose *Magnetic → Transient*
  - Click the OK button
- **Delete the Parametric Analysis**
  - Select and expand Optimetrics
  - Select ParametricSetup1, *RMB → Delete*



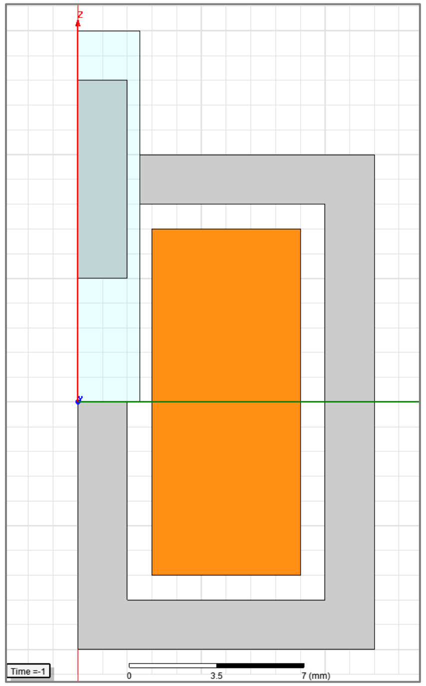
Create Band Object

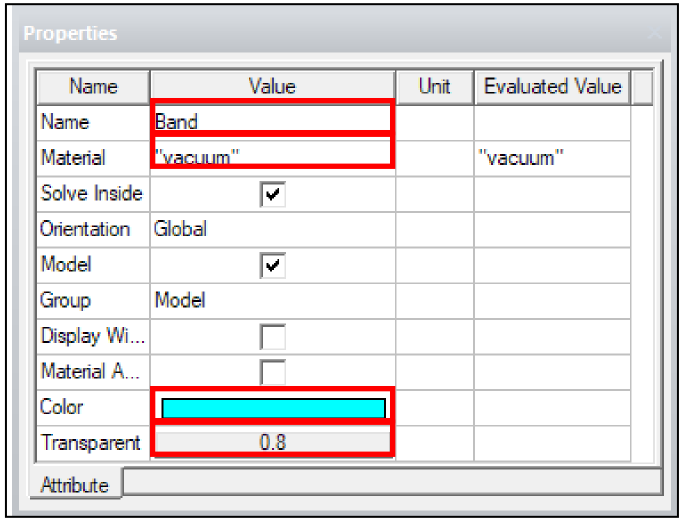
- Create the Band
  - Select the menu item *Draw → Rectangle*
    - Using the coordinate entry fields, enter the position of rectangle

      X: 0, Y: 0, Z: 15, *Press the Enter key*
    - Using the coordinate entry fields, enter the opposite corner

      dX: 2.5, dY: 0, dZ: -15, *Press the Enter key*

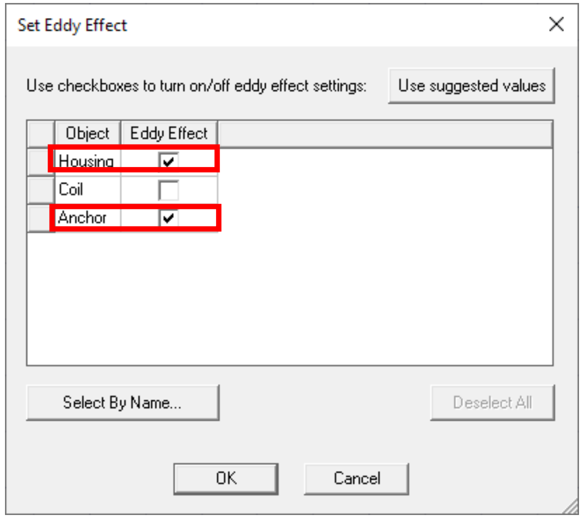
  - Change the name of resulting sheet to Band, the color to light Blue and transparency to 0.8 using the property window
  - Leave the material as it is (Vacuum)

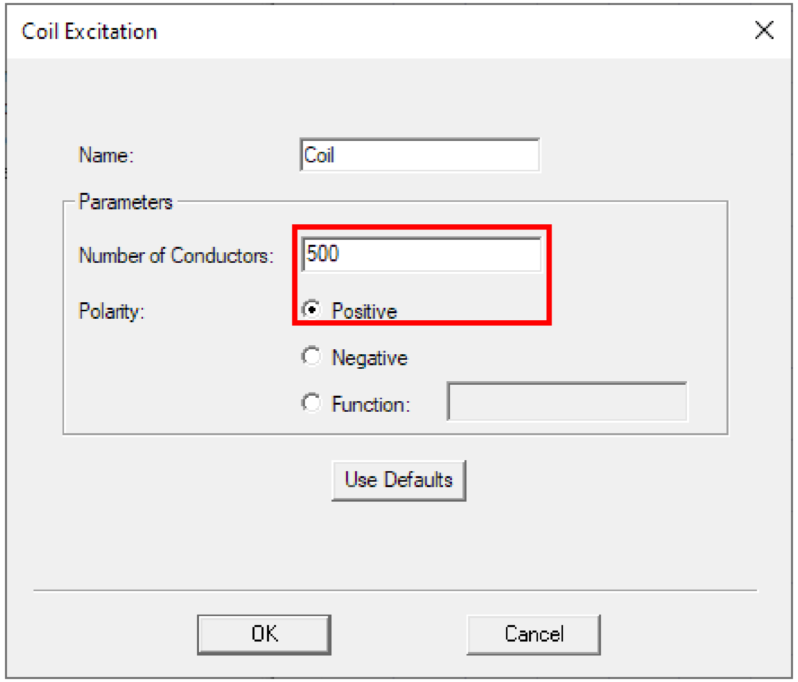




Modify Excitation

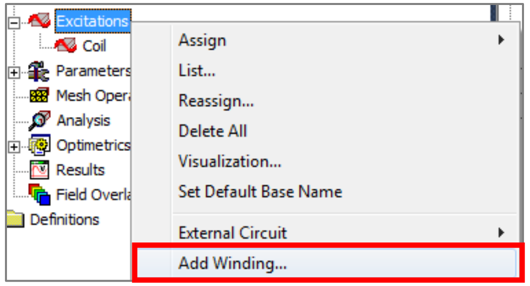
- **Cancel Existing Excitation**
  - Select the Current1 under Excitation and delete it
- **Set Eddy Effects**
  - Select menu item *Maxwell 2D → Excitations → Set Eddy Effects*
  - Housing: ☑ Checked
  - Anchor: ☑ Checked
- **Assign new Excitation**
  - Select the sheet Coil
  - Select menu item *Maxwell 2D → Excitations → Assign → Coil*
  - In Coil Excitation window,
    - Number of Conductors: 500
    - Positive: ☑ Checked
    - Press OK

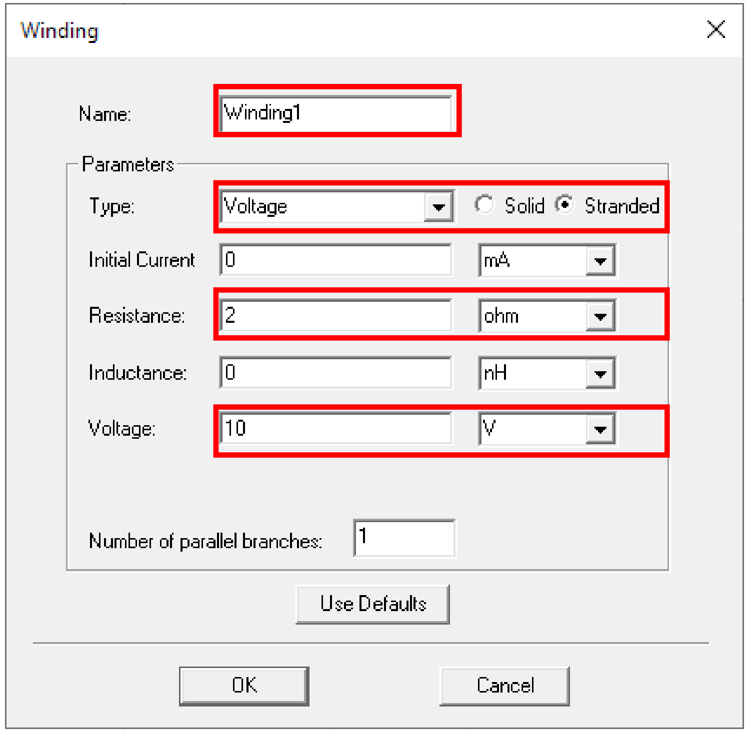


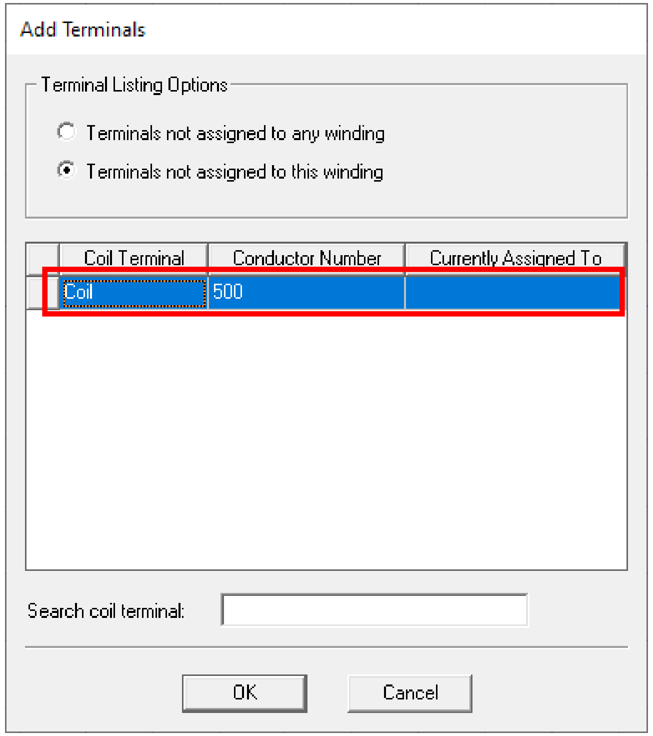


Modify Excitation

- **Modify Excitation – Create windings**
  - Select menu item *Maxwell 2D → Excitations → Add winding*
    - Name: Winding1
    - Type: Voltage
    - Stranded: ☑ Checked
    - Resistance: 2 Ω
    - Voltage: 10 V
    - Press OK
- **Add Coils to Windings**
  - *RMB on Winding1 → Add Coils*
  - Choose Coil
  - Press OK

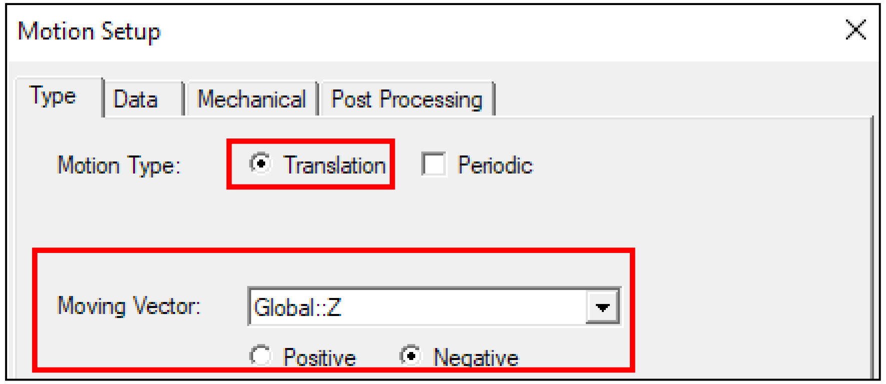


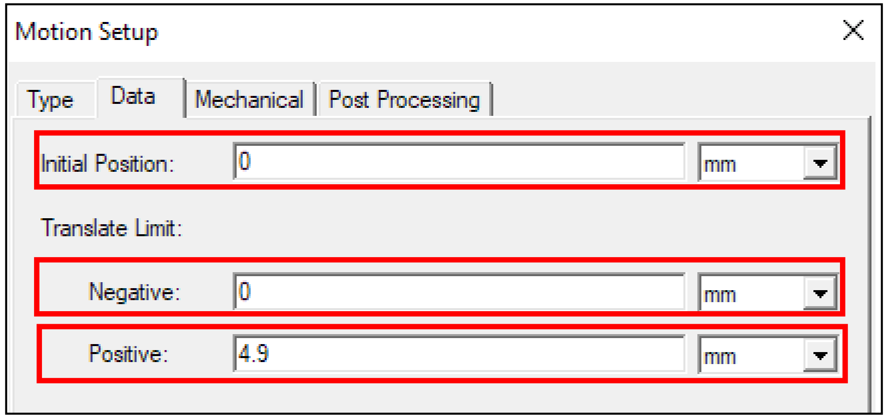




Assign Motion

- **Specify motion**
  - Select the sheet Band from history tree
  - *Select menu item Maxwell 2D → Model → Motion setup → Assign Band*
  - In Motion Setup window,
    - *Type tab*
    - Motion Type: Translation
    - Moving Vector: Global:Z
    - Negative: ☑ Checked
    - *Data tab*
    - Initial Position: 0 mm
    - Negative: 0 mm
    - Positive: 4.9 mm

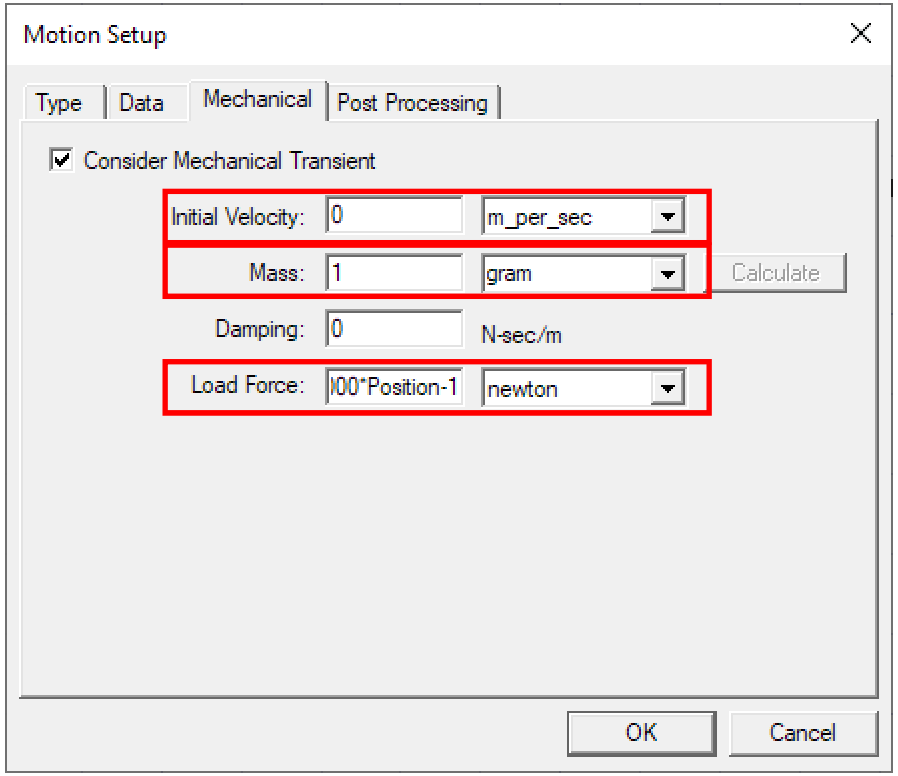




Note: Initial position of 0 mm indicates that motion will start at t = 0 with the Anchor position being as in the geometry. A non-zero initial position (P_0) indicates the simulation would start with Anchor moved by P_0 from the current position

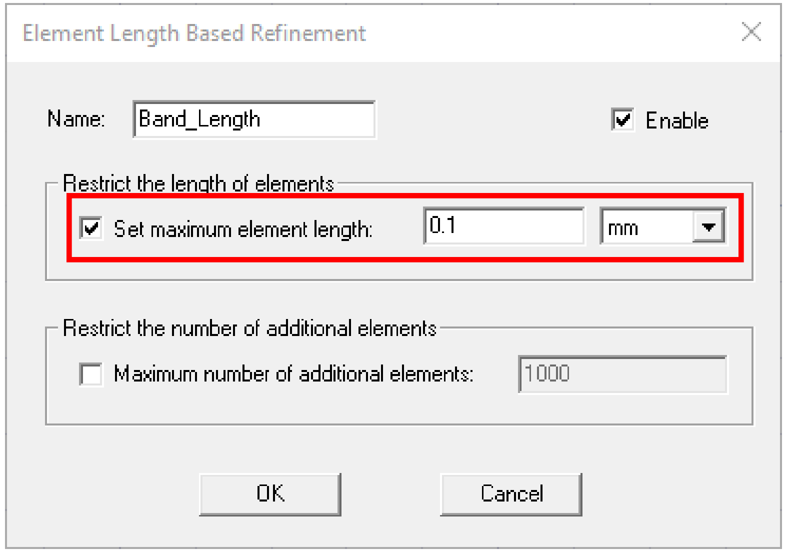
Assign Motion

- *Mechanical tab*
- **Consider Mechanical transient: ☑ Checked**
- **Initial Velocity: 0**
- **Mass: 1 gram**
- **Damping: 0 N-sec/m**
- **Load Force: -1000*Position-1**
- **Press OK**



Mesh Operations

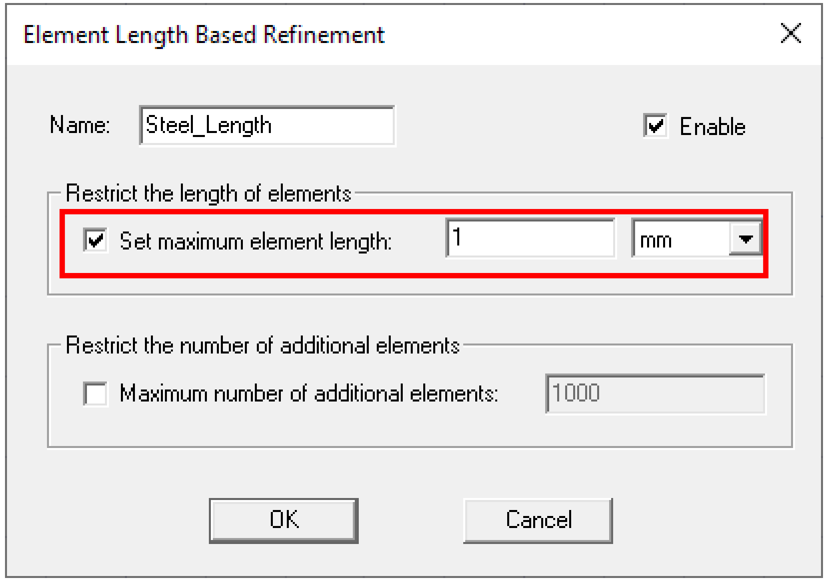
- **Assign Mesh Operations on Band**
  - Select the object Band from the history tree
  - Select menu item *Maxwell 2D → Mesh Operations → Assign → Inside Selection → Length Based*
  - In Element Length Based Refinement window,
    - Name: Band_Length
    - Set maximum element length: ☑ Checked
    - Set maximum element length: 0.1 mm
    - Maximum number of additional elements: ☐ Unchecked
    - Press OK



Note: Meshing is a very critical issue with respect to simulation speed and accuracy. For Force computation, the most critical area is the airgap. Thus, the band mesh is crucial for accurate results.

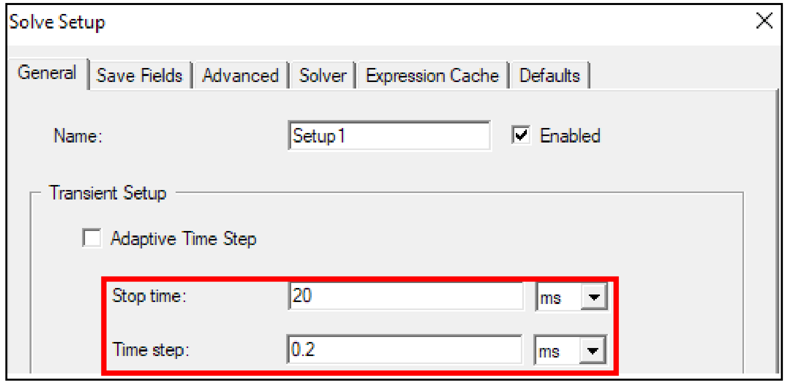
Mesh Operations

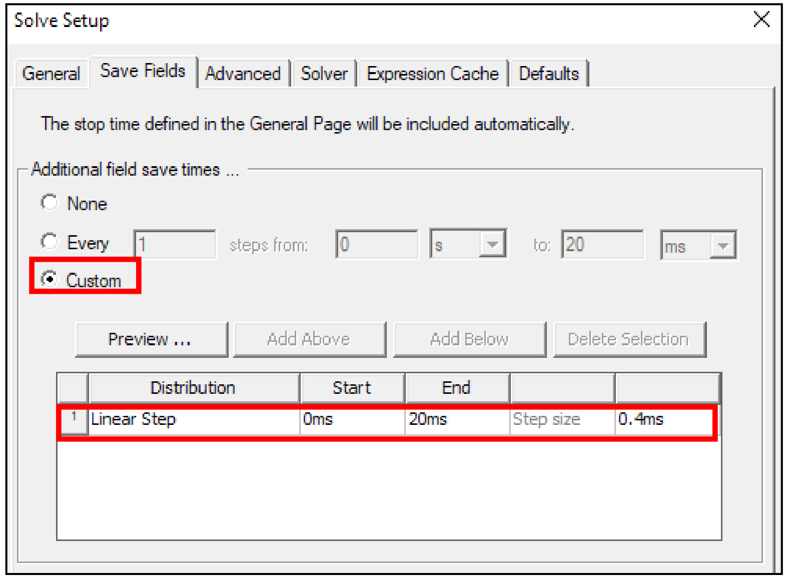
- **Assign Mesh Operations on Housing and Anchor**
  - Select both the sheets Housing and Anchor using the Ctrl key
  - Select menu item *Maxwell 2D → Mesh Operations → Assign → Inside Selection → Length Based*
  - In Element Length Based Refinement window,
    - Name: Steel_Length
    - Set maximum element length: ☑ Checked
    - Set maximum element length: 1 mm
    - Maximum number of additional elements: ☐ Unchecked
    - Press OK

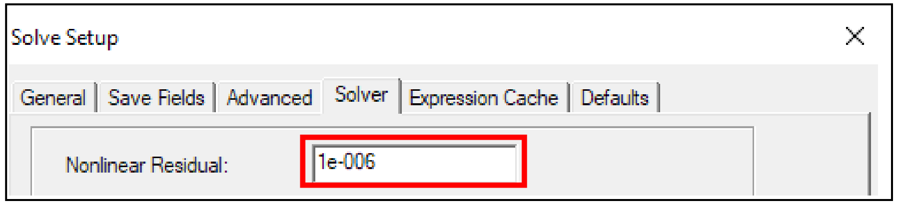


Create Analysis Setup

- Create Analysis Setup
  - Select menu item ***Maxwell 2D→Analysis Setup→Add Solution Setup***
  - In Solve Setup window,
    - General Tab
    - Stop Time: 20 ms
    - Time Step: 0.2 ms
    - Save Fields Tab
    - Select Custom
    - Type: linear Step
    - Start: 0 ms
    - Stop: 20 ms
    - Step: 0.4 ms
    - Press Add to list >>
    - SolverTab
    - NonLinear Residual: 1e-6
    - Press OK



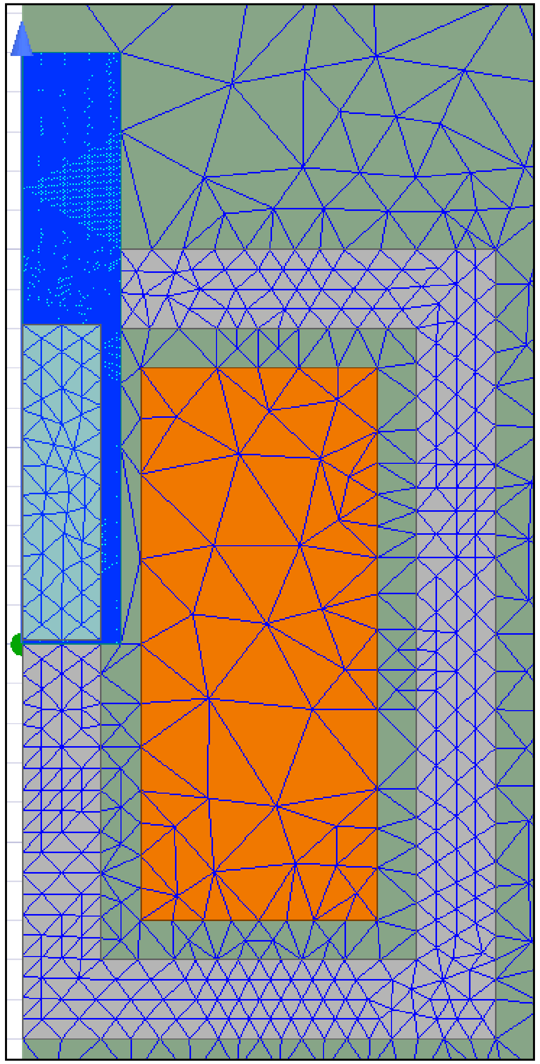




Ansys

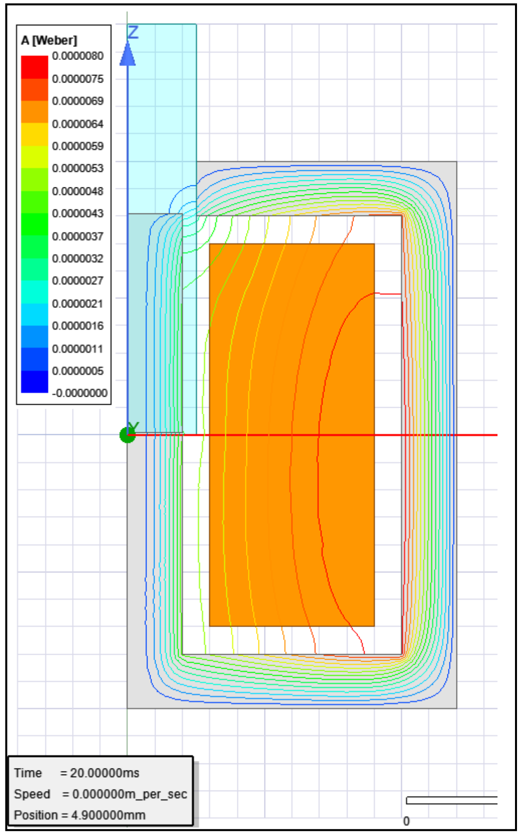
Analyze and Plot Mesh

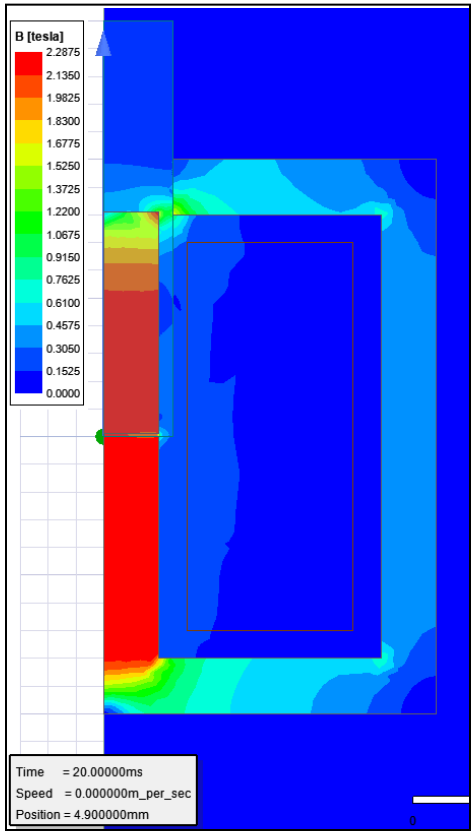
- **Run the Solution**
  - In Project Manager window *RMB on Setup1 → Analyze*
- **Plot Mesh**
  - Select the menu item *View → Set Solution Context*
  - In Set view Context window,
    - Change Time to 0.02 s
    - Press OK
  - Select the menu item *Edit → Select All (Ctrl+A)*
  - Select the menu item *Maxwell 2D → Fields → Plot Mesh*



Results – Create Fields Plot

- Plot Flux_lines
  - Select the menu item *Edit → Select All*
  - *Maxwell 2D → Fields → Fields → A → Flux_Lines*
  - In Create Field Plot window,
    - Press Done
- Plot Mag_B
  - Select the menu item Edit → Select All
  - *Maxwell 2D → Fields → B → Mag_B*
  - In Create Field Plot window,
    - Press Done

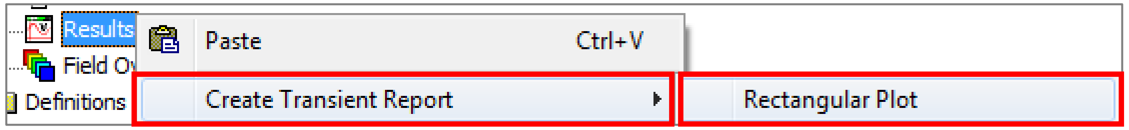


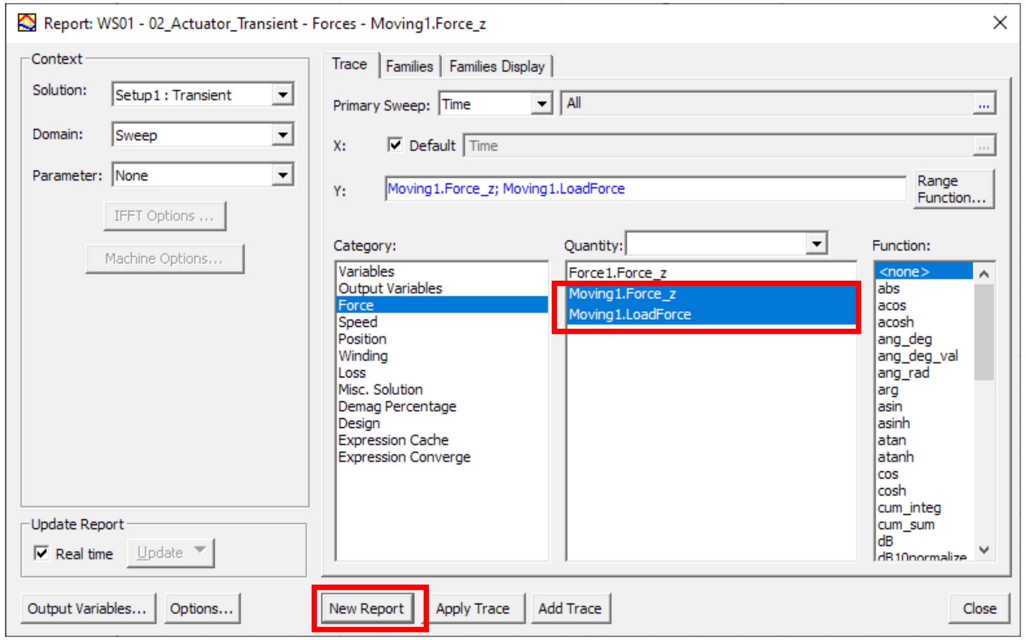


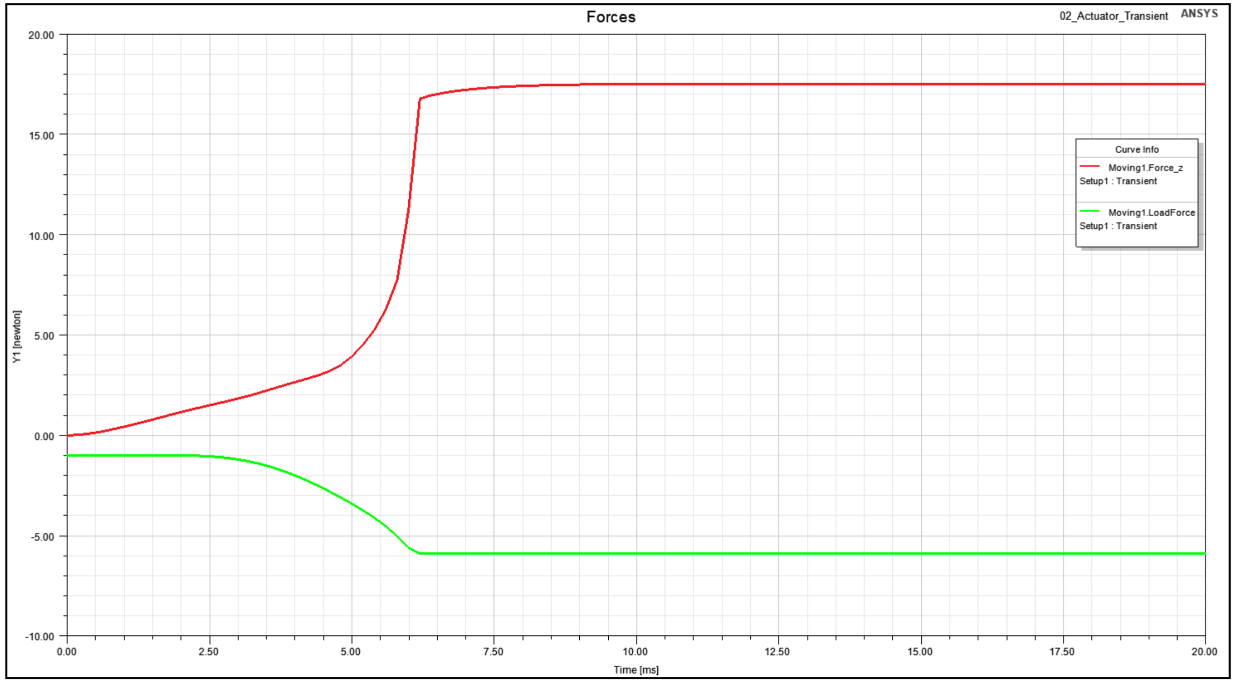
Note: To hide the previously created field plots, right click on plot from Project manager window and uncheck "Plot Visibility"

Plot Forces vs. Time

- Create a XY Plot
  - In Project Manager window *RMB on Results → Create Transient Reports → Rectangular Plot*
  - In Reports window,
    - Category: Force
    - Quantity: Moving1.Force
    - Quantity: Moving1.LoadForce
    - Select New Report

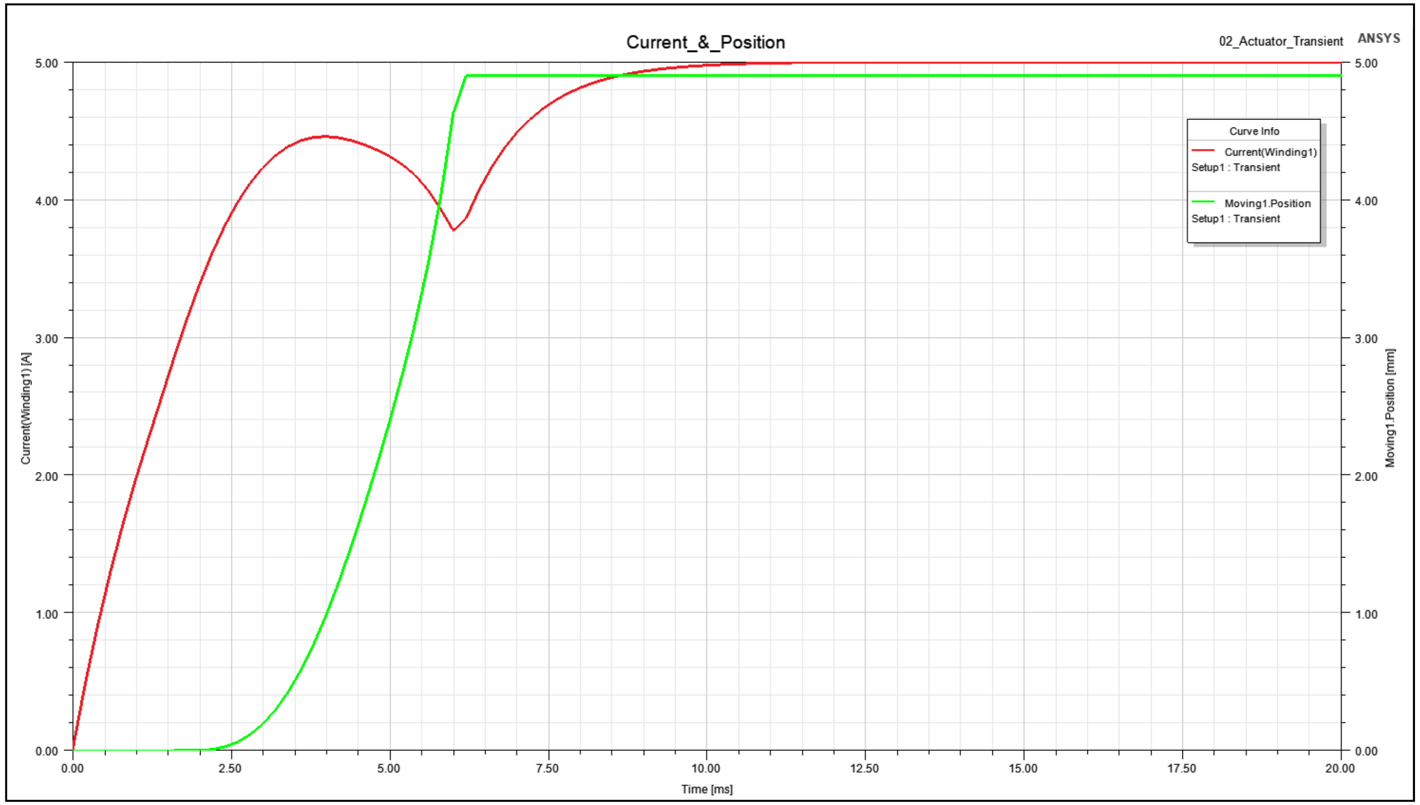






Plot Current and Position vs. Time

- Create a XY Plot
  - In Project Manager window *RMB on Results → Create Transient Reports → Rectangular Plot*
  - In Reports window,
    - Category: Winding
    - Quantity: Current(Winding1)
    - Select New Report
    - Category: Position
    - Quantity: Moving1.Position
    - Add Trace
    - Close



Saving the Project

- **This completes the workshop**
- **Save the file with the name Workshop 01 in the working folder**